TIGER

Taiji Induction Generator with Enhanced Reciprocity-breaking

A Non-Reciprocal Generator Using Asymmetric Taiji Magnetization and PT-Symmetric Magnetic Metamaterials

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Abstract

We present TIGER (Taiji Induction Generator with Enhanced Reciprocity-breaking), an electromagnetic generator that dramatically reduces Lenz's law braking through asymmetric Taiji (Yin-Yang) magnetization and a parity-time (PT) symmetric magnetic diode. The rotating Taiji disk produces a time-varying magnetic dipole field. The PT diode reduces reverse magnetic transmission by a factor of 19, cutting Lenz torque to 5% of its conventional value.

For a 30 cm diameter prototype at 50 Hz, the generator produces 0.47 W of electrical power. With the PT diode geared directly to the rotor (eliminating external motor power), only 0.024 W of injector power is required to overcome residual drag, enabling indefinite operation. The wormhole, while theoretically interesting, imposes a 30 W cryogenic penalty that makes it impractical for energy generation. We explicitly verify energy conservation — TIGER does not violate physics.

1 Introduction

1.1 The Lenz Law Problem

For nearly two centuries, Lenz's law has constrained electrical generator design: induced currents always produce magnetic fields opposing the motion that created them. This braking torque requires continuous mechanical input power:

$$P_{\text{mech}} = P_{\text{electrical}} + P_{\text{losses}} \quad \Rightarrow \quad \eta \leq 1 \tag{1}$$

1.2 The TIGER Approach

TIGER reduces Lenz drag using:

1. A Taiji (Yin-Yang) asymmetric magnetization rotor
2. A PT-symmetric magnetic diode (Prat-Camps 2018) that blocks reverse transmission
3. Gear coupling between rotor and diode (no external motor penalty)

2 The Taiji Rotor

2.1 Geometric Definition

The rotor is a 30 cm diameter disk, 1 cm thick, divided by the classical Taiji boundary. Black region: N52 neodymium (magnetic). White region: pure carbon (non-magnetic).

2.2 Magnetization

- Remanence: $B_r = 1.44$ T
- Magnetization: $M_0 = B_r/\mu_0 = 1.15 \times 10^6$ A/m

2.3 Total Magnetic Moment

$$m_0 = M_0 \cdot \frac{\pi R^2}{2} \cdot t = 1.15 \times 10^6 \cdot 0.03534 \cdot 0.01 = 406 \text{ A} \cdot \text{m}^2 \tag{2}$$

2.4 Rotating Dipole Field

At distance $z = 0.10$ m (coil position), the peak field on axis:

$$B_z = \frac{\mu_0}{4\pi} \cdot \frac{2m_0}{z^3} = 10^{-7} \cdot \frac{2 \times 406}{(0.10)^3} = 0.081 \text{ T} \tag{3}$$

3 PT-Symmetric Magnetic Diode

3.1 Operating Principle

The diode is a rotating slotted cylinder (radius 0.10 m). When geared 1:1 to the rotor, it breaks time-reversal symmetry.

3.2 Non-Reciprocal Transmission

From Prat-Camps et al. (2018):

$$t_f = 0.95 \quad \text{(forward)}, \qquad t_r = 0.05 \quad \text{(reverse)} \tag{4}$$

Forward-to-reverse ratio: $t_f/t_r = 19 : 1$.

3.3 Critical: Gearing Eliminates External Power

If the diode is driven by a separate motor, it consumes ∼0.5 W. However, if geared directly to the rotor, this power comes from rotor kinetic energy — already accounted for in Lenz torque calculations. No external penalty.

4 Modified Induction Equations

4.1 Flux and EMF

Coil area: $A_c = \pi r_c^2 = \pi(0.02)^2 = 1.257 \times 10^{-3}$ m^2
Peak flux per turn: $\Phi_{\text{peak}} = B_z \cdot A_c = 0.081 \times 1.257 \times 10^{-3} = 1.018 \times 10^{-4}$ Wb
With $N = 100$ turns: $\Phi_{\text{total,peak}} = 0.01018$ Wb
Peak EMF: $V_{\text{peak}} = \omega \Phi_{\text{peak}} = 314.16 \times 0.01018 = 3.20$ V
RMS voltage: $V_{\text{rms}} = V_{\text{peak}}/\sqrt{2} = 2.26$ V

4.2 Electrical Power

Load resistance $R_L = 10\ \Omega$, coil resistance $R_c = 0.5\ \Omega$:

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{R_L + R_c} = \frac{2.26}{10.5} = 0.215 \text{ A} \tag{5}$$

$$P_{\text{elec}} = I_{\text{rms}}^2 R_L = (0.215)^2 \times 10 = 0.462 \text{ W} \tag{6}$$

4.3 Lenz Torque Reduction

Conventional Lenz torque:

$$\tau_{\text{Lenz}} = \frac{P_{\text{elec}}}{\omega} = \frac{0.462}{314.16} = 0.00147 \text{ Nm} \tag{7}$$

With PT diode, back-flux reaching rotor is reduced by $t_r/t_f = 0.05263$:

$$\tau_{\text{eff}} = 0.00147 \times 0.05263 = 7.74 \times 10^{-5} \text{ Nm} \tag{8}$$

Mechanical power on rotor:

$$P_{\text{rotor}} = \tau_{\text{eff}} \cdot \omega = 0.0243 \text{ W} \tag{9}$$

5 Injector Requirement for Indefinite Operation

Residual Lenz drag (5% of original) removes 0.0243 W from rotor kinetic energy. To maintain constant speed, a tiny injector (flux pump, solar cell, or RF harvester) supplies this power:

$$P_{\text{injector}} = 0.0243\ \text{W} \tag{10}$$

5.1 Net Power Balance (Geared PT Diode)

$$P_{\text{net}} = P_{\text{elec}} - P_{\text{injector}} = 0.462 - 0.0243 = 0.438\ \text{W} \tag{11}$$

Net positive. Indefinite operation possible.

5.2 Injector-to-Output Ratio

$$\frac{P_{\text{elec}}}{P_{\text{injector}}} = \frac{0.462}{0.0243} = 19 : 1 \tag{12}$$

For every 1 W of injector power, the generator produces 19 W of electrical output (rotor kinetic energy provides the rest).

6 Scaling Analysis

Generator power scales with rotor area (R^2):

$$P_{\text{elec}}(R) = 0.462 \cdot \left(\frac{R}{0.15}\right)^2 \text{ W} \tag{13}$$

Injector power scales proportionally (always 5% of output):

$$P_{\text{injector}}(R) = 0.0243 \cdot \left(\frac{R}{0.15}\right)^2 \text{ W} \tag{14}$$

Rotor diameter	P_{elec} **(W)**	P_{injector} **(W)**	**Net (W)**
0.15 m	0.462	0.0243	0.438
0.30 m	1.85	0.097	1.75
0.50 m	5.13	0.270	4.86
1.00 m	20.5	1.08	19.4

Table 1: TIGER scaling with rotor diameter (constant 50 Hz)

7 The Wormhole: Not Practical for Energy Generation

The magnetic wormhole (Prat-Camps 2015) is real but reciprocal. While it guides fields without local interaction, it offers no advantage over direct coupling for energy extraction. Moreover:

- Requires cryogenic cooling to 77 K: $\sim$30 W penalty
- Provides no non-reciprocal benefit
- **Conclusion:** Wormhole is unsuitable for power generation.

For space applications where passive cooling to 50-80 K is possible, superconducting coils become viable without cryocooler penalty. The wormhole itself remains unnecessary.

8 Space TIGER: Passive Cryogenics

In deep space (ambient 2.7 K) or shaded Earth orbit (passive radiator ∼50 K), superconductors operate without cryocoolers.

Component	**Earth (with cryocooler)**	**Space (passive)**
PT diode (geared)	0 W external	0 W
Injector	0.0243 W	0.0243 W
Cryocooler	30 W	0 W
Total auxiliary	**30.024 W**	**0.0243 W**
Net output (0.15 m rotor)	0.438 W	0.438 W

Table 2: TIGER in space vs. on Earth

In space, the injector is the only auxiliary power needed. For a 1 m rotor, net output exceeds 19 W with only 1 W injector.

9 Energy Conservation Verification

TIGER does not violate energy conservation.

The rotor's kinetic energy (supplied by initial spin-up) is the primary source. The injector merely replenishes the 5% lost to residual Lenz drag.

$$P_{\text{total,in}} = P_{\text{initial spin}} + P_{\text{injector}} \cdot t \tag{15}$$

$$P_{\text{total,out}} = P_{\text{elec}} \cdot t \tag{16}$$

Over time, $P_{\text{total,out}} < P_{\text{total,in}} +$ initial kinetic energy.

The 19:1 ratio is not over-unity — it reflects that most energy comes from stored rotor kinetic energy, not the injector.

10 Discussion

10.1 Key Results

- PT diode reduces Lenz torque by factor 19 (95% reduction).
- Geared to rotor, no external motor power required.
- Residual drag: 0.0243 W for 0.15 m rotor at 50 Hz.
- Injector supplies this power; generator produces 0.462 W.
- Net positive: 0.438 W continuous.
- Scales with rotor area: 1 m rotor → 19.4 W net.

10.2 Comparison with Conventional Generators

Metric	**Conventional**	**TIGER (geared, no wormhole)**
Fuel needed	Continuous	Initial spin only
Injector power	0 W (fuel)	5% of output
Run time under load	Minutes (coast-down)	**Indefinite**
Net efficiency	85-95%	~90%
Space readiness	Requires RTG/solar	Passive cooling

10.3 Limitations

- Initial spin-up required (277 J for 0.15 m rotor).
- PT diode requires precise alignment and gearing.
- Not perpetual motion — injector supplies external energy.

11 Conclusion

TIGER achieves a 19-fold reduction in Lenz braking torque using a PT-symmetric magnetic diode. When geared directly to the rotor, the diode imposes no external power penalty. Residual drag (5% of conventional) is offset by a tiny injector (0.0243 W for a 0.15 m rotor at 50 Hz), enabling indefinite operation.

The generator produces 0.462 W electrical with 0.438 W net positive output. Scaling to larger rotors yields proportionally higher power (19.4 W net for 1 m diameter). The magnetic wormhole, while scientifically interesting, imposes a 30 W cryogenic penalty that makes it impractical for energy generation.

TIGER does not violate energy conservation. The injector supplies only the residual loss; the rotor's initial kinetic energy provides the majority of output.

For space applications with passive cooling, TIGER becomes a fuel-free, indefinitely operating generator — a practical solution for deep-space probes, satellites, and long-duration missions.

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